Channel Adaptive and Sparsity Personalized Federated Learning for Privacy Protection in Smart Healthcare Systems

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***Abstract*—With the booming development of Smart Healthcare Systems (SHSs), employing federated learning (FL) in SHS devices has become a research hotspot. FL, as a distributed learning framework, can train models without sharing the original data among users, and then protect the user privacy. Existing research has proposed many methods to improve the security and efficiency of FL, which may not fully consider the characteristics of SHSs. Specifically, the requirements of privacy protection and efficiency pose significant challenges to FL. Current studies have struggled to balance privacy security and efficiency, and the degradation of model training efficiency in SHSs can be critical to patient health. Therefore, to improve the privacy protection of healthcare data and ensure communication efficiency, this work proposes a novel personalized FL framework based on Communication quality and Adaptive Sparsification (pFedCAS). In order to achieve privacy protection, a control unit is proposed and introduced to adjust the sparsity of the local model adaptively. To further improve the training efficiency, a selection unit is added during global model aggregation to select suitable clients for parameter updates. Finally, we validate the proposed method operated on the HAM10000 dataset. Simulation results validate that pFedCAS can not only improve privacy protection, but also gain an improvement of 15% in training accuracy and a reduction of 30% in training costs based on communication quality. The simulation results also validate the excellent robustness of pFedCAS to non-iid data.**

***Index Terms*—Adaptive sparse, efficiency, personalized federated learning, privacy protection, SHS.**

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I. Introduction

In RECENT years, the booming development of Smart Healthcare Systems (SHSs) has brought significant changes to the healthcare industry and significantly improved the quality of life for humans. The SHSs transform traditional healthcare services into personalized, precise, and efficient services, providing patients with higher-quality healthcare experience [1]. In smart medical systems, smart wearable devices and other SHS devices can monitor users' health status in real-time and collect healthcare data. These massive data can be uploaded synchronously to explainable artificial intelligence (XAI) supported frameworks to achieve efficient smart healthcare applications, such as remote health monitoring, disease prediction, and medical image processing [2], [3].

Traditional smart healthcare systems usually adopt a centralized approach and framework for artificial intelligence (AI) [4]. Specifically, various SHS devices upload collected healthcare data to a data center, which is then processed and analyzed by the central AI system [5], [6]. However, with the rapid development of SHSs and the explosive growth of healthcare data, there is an increased risk of data security. Uploading raw healthcare data directly to a central server can easily lead to user privacy or data leakage, which may pose a significant threat to user privacy and data security [7]. In addition, the transmission and processing of massive raw healthcare data usually cannot ensure high timeliness, which may result in abnormal data of patients not being processed in a timely manner, which can be fatal for patients [8]. Therefore, compared with a centralized AI framework, a distributed XAI framework is more suitable for smart healthcare systems that require privacy protection and high scalability.

Federated learning (FL), as an emerging distributed learning framework, can potentially ensure privacy protection and improve the efficiency of healthcare data processing [9], [10]. Under a specialized mechanism incorporating homomorphic encryption, FL protects user data privacy by preserving confidentiality during the exchange of model parameters [11], [12]. Specifically, FL is a distributed XAI framework that trains a

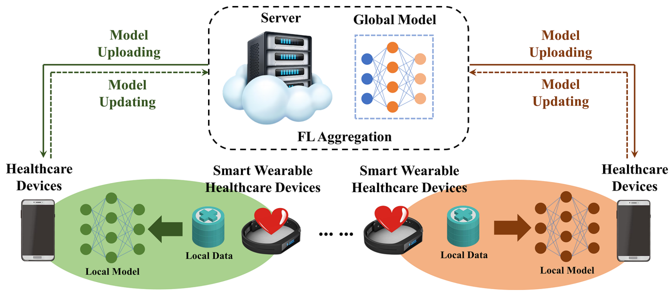


Fig. 1. FL-based smart healthcare architecture.

global model by aggregating local model parameters updated at multiple healthcare data clients without directly accessing user data. A typical architecture of FL-based smart healthcare is shown in Fig. 1, which provides a paradigm for privacy-focused frameworks in SHSs. Such a scheme greatly improves the efficiency and quality of data analysis and processing, which makes FL highly suitable for SHSs with extremely high requirements for timeliness and privacy protection.

A. Related Works

Recently, new application scenarios and SHS requirements have brought many challenges and difficulties to FL. Firstly, the security of healthcare sensor data and the requirement for data processing efficiency pose significant challenges to the efficient FL framework of SHSs. Xu et al. [13] proposed the FL method with a Minimum Square Quantization Error (FedMSQE), which reduced the network scale by quantizing the size of neural networks. It can improve the security of data transmission and reduce the communication cost. Jiang et al. [14] proposed a new federated edge learning framework based on hybrid differential privacy and adaptive compression. The process began with the implementation of adaptive gradient compression readiness, followed by creating an industrial FL model. Finally, an adaptive differential privacy model is utilized to optimize the entire process, ensuring the privacy protection of gradient transmission. Sun et al. [15] proposed a scalable and lightweight FL classification system for healthcare data. This efficient system took a small storage space and little training time. Furthermore, it was easy to extend to new classes and transfer to new tasks. Dieuleveut et al. [16] proposed a communication-efficient FL, named FedEM, which handled partial participation of local models, and was robust to the data heterogeneity. Meanwhile, FedEM compressed appropriately defined complete data with sufficient statistics to reduce communication costs.

Secondly, the data from different SHS devices are usually heterogeneous across clients, which may cause the performance degradation of FL model training. Ullah et al. [17] implemented a scalable FL framework for interactive smart healthcare systems with intermittent clients, operated on chest X-ray images. The data augmentation method was used to balance datasets for the training of local models. Zhang et al. [18] proposed a heterogeneity-aware FL method, named SplitAVG, to alleviate the performance degradation resulting from data heterogeneity in the FL framework. SplitAVG leveraged the simple network split and feature map concatenation strategies to encourage the local model to train an unbiased estimator of the target data distribution. Additionally, SplitAVG can be adapted to different base convolutional neural networks and generalized to various healthcare imaging tasks. Khowaja et al. [19] proposed SelfFed framework work for SHSs, consisting of two stages. The first stage is a pre-training paradigm that utilizes a decentralized approach to perform enhanced modeling using a Swin Transformer-based encoder. The first stage was used to overcome data heterogeneity. The second stage was a fine-tuning paradigm introducing a contrastive network and a novel aggregation strategy, which addressed the problem of label scarcity. T Dinh et al. [20] proposed a personalized FL method, named pFedMe, applying Moreau envelopes as regularized loss functions of local models, which helped decouple personalized model optimization from the global model learning in a bi-level problem stylized for personalized FL. Advanced explorations for FL have been made to improve the privacy protection of healthcare data and reduce communication costs. However, they have not fully considered the characteristics of SHSs. Many existing studies have attempted to reduce communication and computation costs through methods such as sparsification, quantization, or compression, which may be fixed or do not consider different resource limitations of clients. Although the aforementioned methods can significantly improve communication efficiency, they may lead to performance degradation. More importantly, the SHS differs from other application scenarios, where even slight differences in model performance can be critical to patient safety and health. Therefore, SHS requires a network framework that can achieve privacy protection while maximizing model accuracy under specific communication conditions.

B. Research Contributions

To address the aforementioned issues, this work proposes a new personalized FL framework based on Communication quality and Adaptive Sparsification, named pFedCAS. Specifically, our main contributions can be summarized as follows:

- Communication quality factors are defined to characterize the communication status of the clients. The integration of data heterogeneity and communication status constraints heterogeneity yields an efficient optimization problem. Furthermore, we make assumptions and provide proof for the rationality of this optimization problem.
- A new communication-quality-based local model training scheme consisting of two components is designed. The main component is responsible for local model training, and the auxiliary component, which is the sparsity control unit, controls the sparsity of the local model according to the communication quality. This framework can achieve privacy protection and maximize model performance under the constraint of communication quality. For researchers, the sparse control units enable model sparsity by assigning different values to parameter updates, which can demonstrate the contributions of each participant to the final model. This increases transparency, allowing

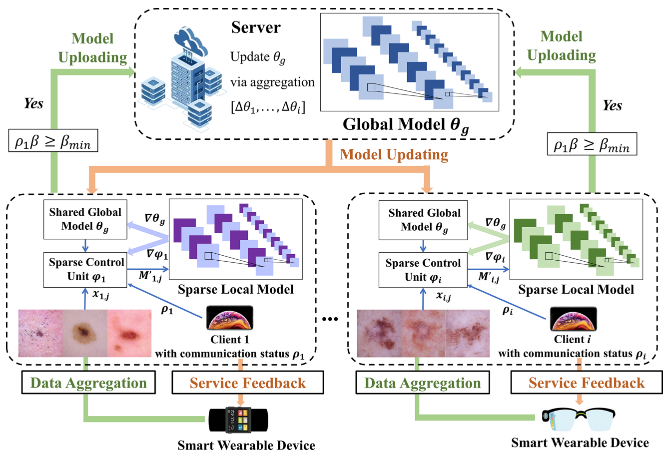


Fig. 2. pFedCAS-based data aggregation and model update.

stakeholders to understand the impact of each participant on the model, thereby enhancing the interpretability and visualizability of the model aggregation process.

- We design a new personalized FL framework based on communication quality and adaptive sparsification for SHSs. During the global model aggregation process, a selection unit is added to filter clients with high communication quality and significant contributions to global updates, thereby efficiently training of the entire system. Simulation results validate that the proposed framework can improve communication efficiency while ensuring privacy protection.

The structure of this paper is as follows: Section II details the modeling of the system model and optimization problem. Section III provides a detailed description of the proposed method's specific framework and algorithm process. Section IV validates the proposed method through simulation experiments. Finally, Section V is a conclusion of the entire paper.

II. Framework Overview

This work aims to improve the privacy protection while ensuring communication efficiency utilizing the communication-quality-based adaptive sparsity method. This section provides a detailed description of the proposed personalized federated learning system model. Furthermore, the optimization problem is derived by incorporating data heterogeneity and communication resource heterogeneity.

A. System Model

Fig. 2 illustrates the framework overview of the proposed personalized FL based on communication-quality adaptive sparsity for SHSs. In the framework of pFedCAS, smart wearable devices collect user data, which is then uploaded to clients such as smartphones with inference and computing capabilities. The set of clients is denoted by $\mathcal{N} = \{1, 2 \ldots, N\}$. Meanwhile, each client $i \in \mathcal{N}$ is considered to have its own private dataset $\mathcal{G}_i$ sampled from a local distribution $\mathcal{D}_i$ over $\mathcal{X} \times \mathcal{Y}$, which is heterogeneous. Meanwhile, a global data distribution $\mathcal{D}_g$ is defined as a mixture of $\mathcal{D}_i$. When client i finishes its data aggregation, the aggregated data is used to train local model $h_{\theta_i} \in \mathcal{H}$, which is parameterized as θ_i. After the training is completed, client i uploads sparse $\Delta\theta_i$ to the server. Then the server updates the global model h_{θ_g} via aggregating $\{\Delta\theta_{g,i}\}_{i\in\mathcal{N}}$ and returns the updated parameters to each client. LeNet [21] is applied as the neural network of the global model, whose architecture is shown in Fig. 3. Besides, we consider that clients have different constraints of communication resources. According to the Shannon's formula, the instantaneous rate for client i is formulated as

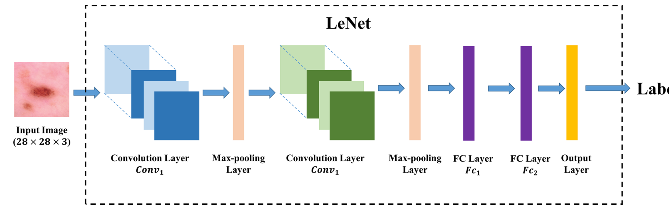


Fig. 3. Architecture of LeNet.

$$R_i = W_i \log_2 \left(1 + \frac{P_i g_i}{N_0 W_i}\right), \;\; \forall i \in \mathcal{N}, \tag{1}$$

where W_i is the bandwidth occupied by client i, and N_0 denote the noise spectral density. P_i and g_i denotes the transmission power and the channel gain of client i, respectively.

Furthermore, we define a standard data rate R_{std}. If the data rate of client i is greater than or equal to R_{std}, local model sparsification is not necessary; otherwise, model sparsification is required. Meanwhile, a parameter ρ_i is defined to represent the communication quality of client i, which is expressed as

$$\rho_i(t) = \begin{cases} 1, & R_i(t) \geq R_{std}, \\ \frac{R_i(t)}{R_{std}}, & R_i(t) < R_{std}. \end{cases} \tag{2}$$

Through the selection unit applying communication quality factor ρ_i, clients can be selected during the global model aggregation process. We define the set $\mathcal{N}_k$ as the set of clients suitable for parameter updates. By selecting clients to upload in this way, the proposed method can further improve the efficiency of model training.

B. Problem Formulation

We take into account the data heterogeneity and resources constraints heterogeneity more effectively by optimizing the following objective [22], which can be expressed as

$$\min_{\{\theta_i\}} \sum_i p_i \mathbb{E}_{(x,y)\sim\mathcal{D}_i} \left[f(\theta_i; x, y)\right], \tag{3a}$$

$$\text{s.t. size}(\theta_i) \leq \rho_i d, \;\; \forall i \in \mathcal{N}, \tag{3b}$$

where $f(\theta_i; x, y) \triangleq l(h_{\theta_i}(x), y)$ and $l(h_{\theta_i}(x), y)$ is the loss function. Size(θ_i) denotes the number of parameters of θ_i and d is the dimension of the global model parameters. $p_i \in (0, 1]$ is aggregation weight of client i, which is defined as [23]

$$p_i = \frac{|\mathcal{G}_i|}{\sum_i |\mathcal{G}_i|}. \tag{4}$$

To reduce the performance degradation caused by the compression and lower the computational training cost, we jointly learn to sparse and train personalized local models. Therefore, Problem (3) can be reformulated as the following problem:

$$\min_{h_{\theta_i}} \sum_i p_i \mathbb{E}_{(x,y)\sim\mathcal{D}_i} \left[l(h_{\theta_i}(x), y) \right], \tag{5a}$$

$$\text{s.t. } \mathbf{Count}_{\text{ nonzero}}(\theta_i) \leq \rho_i d, \ \forall i \in \mathcal{N}, \tag{5b}$$

where $\text{Count}_{\mathbf{nonzero}}$ is used to count the number of non-zero elements. Due to the data heterogeneity and resources constraints heterogeneity, if no assumptions are made about the data distribution of different clients, optimizing parameter θ_i directly may lead to difficulty in convergence. Therefore, the following assumptions is needed to ensure the learnability and convergence of the learning process:

Assumption 2.1: Denote $h_{\theta_i^*}$ and $h_{\theta_g^*}$ as the optimal estimator for data distributions $\mathcal{D}_g$ and $\mathcal{D}_i$, respectively. There exists a $\mathcal{D}_g$ such that the variance between the local and global gradients is bounded:

$$\left\| \nabla_\theta f(\theta_g^*) - \nabla_\theta f(\theta_i^*) \right\| \leq \sigma_i, \tag{6}$$

where $f(\cdot)$ is compact notation and f is μ-strongly convex.

Proposition 2.1: According to Assumption 2.1, there exist a d-dimensional continuous weight vector $M_i^* \in \mathbb{R}^d$ such that $\theta_i^* = M_i^* \circ \theta_g^*$, where $\circ$ denotes Hadamard production over model parameters of θ. M_i^* can be bounded as:

$$\left\| (M_i^* - 1)^{\circ 2} \right\| \leq \frac{A_i \sigma_i^2}{\mu^2 \left\| (\theta_g^*)^{\circ 2} \right\|}, \tag{7}$$

$$A_i = \frac{1}{2} \left[\frac{(Z_i - z_g) Z_g^2}{(Z_g - z_i) z_g^2} + \frac{(Z_g - z_i) z_g^2}{(Z_i - z_g) Z_g^2} \right], \tag{8}$$

where $\circ 2$ denotes Hadamard power of 2, z_i and z_g are the infimum of θ_i and θ_g, respectively, Z_i and Z_g are the supremum of θ_i and θ_g, respectively.

Proof 2.1: See Appendix.

According to Proposition 2.1, Problem (5) can be optimized via generating local sparse models parameterized by $\theta_i' = M_i \circ \theta_g$. Meanwhile, $\beta \in (0, 1]$ is defined as the model sparsity constraint of the whole system. Then, Problem (3) can be transformed into the following form:

$$\min_{\theta_g, \{M_i\}} \sum_i p_i \mathbb{E}_{(x,y)\sim\mathcal{D}_i} \left[f(M_i \circ \theta_g; x, y) \right], \tag{9a}$$

$$\text{s.t. } \mathbf{Count}_{\text{ nonzero}}(M_i) \leq \rho_i d\beta, \ \forall i \in \mathcal{N}. \tag{9b}$$

Furthermore, we obtain a control weight vector M_i to implement local models that can adaptively adjust the sparsity level based on the communication quality of client i. Moreover, communication quality factor ρ_i directly affects the sparsity level of the local model, thus indirectly affecting the contribution of the local model to the global model updating. Specifically, when a client's communication quality is poor, the sparsity level of its local model increases. Excessive sparsity can lead to the local model making almost no contribution to the global model update. Therefore, $\beta_{\min}$ is defined as minimum sparsity constraint to filter out clients that are suitable for uploading parameter updates, thus further improving the training efficiency of the entire system. Then, we can obtain the final optimization problem, which can be expressed as

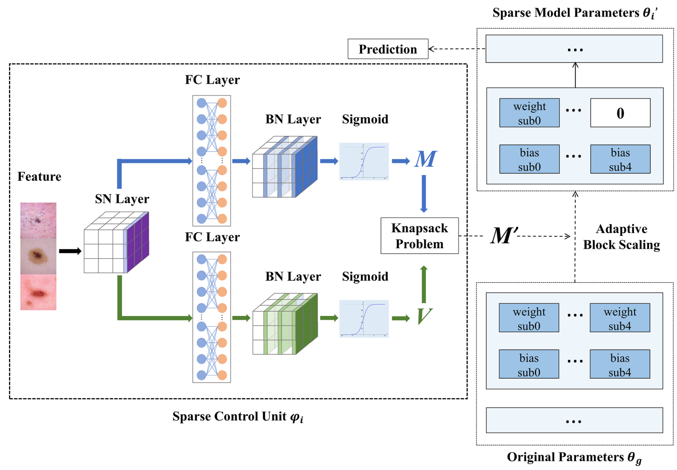


Fig. 4. Framework of sparse control unit in local training.

$$\min_{\theta_g, \{M_i\}} \sum_i p_i \mathbb{E}_{(x,y)\sim\mathcal{D}_i} \left[f(M_i \circ \theta_g; x, y) \right], \tag{10a}$$

$$\text{s.t. } \mathbf{Count}_{\text{ nonzero}}(M_i) \leq \rho_i d\beta, \tag{10b}$$

$$\rho_i \beta \geq \beta_{\min}, \ \forall i \in \mathcal{N}. \tag{10c}$$

III. Communication-Quality-Based Adaptive Sparsity Scheme

To address the optimization problem (10) presented in Section II, we have designed an adaptive sparse control unit. This unit sparsifies the local models by adaptively generating a weight vector M. This section provides a detailed description of the architecture of the sparse control unit and presents the specific workflow of the proposed algorithm.

A. Sparse Control Unit

Utilizing control weight M_i proposed in Section II, a sparse control unit is constructed to achieve personalized local model sparsification. The specific framework of the sparse control unit is shown in Fig. 4. Moreover, further to improve the training efficiency of the local sparse model, M_i is blocked into $M_{i,j}'$ at the batch level, which can be expressed as

$$M_{i,j}' = v_{\varphi_i}(\mathbf{x}_{i,j}), \ \forall i \in \mathcal{N}, \ \forall j \in \mathcal{B}, \tag{11}$$

where φ_i denotes the parameters of the personalized control unit, and $\mathcal{B} = \{1, 2, \ldots, B\}$ is a set of data batches. Meanwhile, B is the number of sub-blocks and d_x is the dimension of the input feature x. $v_{\varphi_i} : \mathbb{R}^{d_x} \to \mathbb{R}^B$ represents using data batch $\mathbf{x}_{i,j}$ and communication quality ρ_i to predict and generate block weights

Algorithm 1: Train Sparse Control Unit of Client i in a Step.

```
Input: Feature x_i, learning rate η_c,
for j = 1 → B do
    Get handled data batches:
    x̂_{i,j} = SwitchNorm(x_{i,j}),
    V̂_{i,j} = FC_V(x̂_{i,j}), M̂_{i,j} = FC_M(x̂_{i,j}),
    Ṽ_{i,j} = BatchNorm(V̂_{i,j}),
    M̃_{i,j} = BatchNorm(M̂_{i,j}),
    V_{i,j} = Sigmoid(Ṽ_{i,j}), M_{i,j} = Sigmoid(M̃_{i,j}),
    Solve the Knapsack Problem (13) with V_{i,j}, M_{i,j},
    and obtain the binary vector U,
    Get sparse control weight: M'_{i,j} = U ∘ M_{i,j},
    Get personalized sparse local model:
    θ̂_{i,j} = θ_g ⊚ M'_{i,j},
    Calculate the loss function: f(θ̂_{i,j}; x_{i,j}, y_{i,j}),
    Update φ_i = φ_i − η_c ∇_φ θ̂_{i,j} ∇_θ f(θ̂_{i,j}; x_{i,j}, y_{i,j}).
end for
```

$M'_{i,j}$. Then, we can obtain the parameters of personalized adaptive sparse local model, which can be expressed as

$$\hat{\theta}_{i,j} = \theta_g \circledcirc M'_{i,j} = v_{\varphi_i}(\mathbf{x}_{i,j}),\ \forall i \in \mathcal{N},\ \forall j \in \mathcal{B}, \tag{12}$$

where $\circledcirc$ represents block-based generation of parameters θ_g.

As shown in Fig. 4, the control unit consists of two paths. The generation path is used to generate the initial control weight vector M, which may violate the sparsity constraints caused by communication quality. The control path adjusts the weights M into a sparse vector M' the sparsity constraint by generating block values V. The structure of two paths is basically the same, with two fully connected layers of the same size, both of which are Bd_x. Therefore, the control unit parameter φ_i, as a mixture of the parameters of two fully connected layers, has a size of $d_\varphi = 2Bd_x$. A complete training process of the sparse control unit is shown in Algorithm 1.

To maximize the efficiency of parameter updates while satisfying the sparsity constraint, we solve the knapsack problem to obtain the optimal block control weights M' that meet the constraint. Specifically, $L \in \mathbb{Z}^B$ is defined as a lookup vector of the sub-blocks parameter, and a binary vector $U \in \{0,1\}^B$ is defined to filter the required parameters from weight M. Then, maximize block values while satisfying the sparsity constraint by addressing the following problem:

$$\max_U\ U \cdot V, \tag{13a}$$

$$\text{s.t. } U^\top \cdot L \leq \rho_i d\beta. \tag{13b}$$

In addition, the adaptive sparse control unit in local model training not only significantly reduces the training cost but also further enhances the privacy protection of the federated learning framework. Firstly, our proposed sparse control units synchronize the learning of model sparsity with local model parameter updates, avoiding the leakage of local data. Secondly, model sparsity acts as a mask layer added to the uploaded parameters of the local model, making it more challenging to access the parameter space of the local model. Lastly, the presence of unknown personalized sparse control units further reduces the risk of data leakage.

B. pFedCAS Algorithm

As shown in Fig. 2, each client is composed of two components. The main component is used to train the local model. The auxiliary component, which is the sparse control unit, is used to learn and generate control weights M to control the sparsification of the local model. It should be noted that these two learning processes are parallel. In addition, according to the constraints in (10), a decision unit is added to filter out local models suitable for uploading model updates. The specific algorithm process is shown in Algorithm 2.

Firstly, we analyze the generalization performance of pFedCAS. Let $\mathcal{L}(\theta_g, \varphi_i)$ and $\tilde{\mathcal{L}}(\theta_g, \varphi_i)$ represent the expected loss and empirical loss, respectively, which are expressed as

$$\mathcal{L}(\theta_g, \varphi_i) = \mathbb{E}_{(x,y)\sim\mathcal{D}_i} f[(v_{\varphi_i(x)} \circledcirc \theta_g); x, y], \tag{14}$$

$$\tilde{\mathcal{L}}(\theta_g, \varphi_i) = \frac{1}{|\mathcal{G}_i|} = \sum_{(x,y)\in\mathcal{G}_i} f[(v_{\varphi_i(x)} \circledcirc \theta_g); x, y]. \tag{15}$$

Assumption 3.1: The parameters θ_g and φ_i are bounded within a spherical space of radius R: $\|\theta_g\| \leq R$, $\|\varphi_i\| \leq R$. There exists the following Lipschitz conditions: $|f(\theta; x, y) - f(\theta'; x, y)| \leq L_f \|\theta - \theta'\|, |h_\theta - h_{\theta'}| \leq L_h \|\theta - \theta'\|, |v_{\varphi_i}(x) - v_{\varphi'_i}(x)| \leq L_v \|\theta - \theta'\|$.

Theorem 3.1: According to Assumption 3.1, with probability at least $1 - \delta$, there exists $\tilde{S}_i$ such that when the local data size $\mathcal{G}_i$ satisfies $|\mathcal{G}_i| \geq \tilde{S}_i$, the following inequality holds:

$$\left| \mathcal{L}(\theta_g, \varphi_i) - \tilde{\mathcal{L}}(\theta_g, \varphi_i) \right| \leq \epsilon, \forall \theta_g, \varphi_i. \tag{16}$$

The lower bound on the local data size is formulated as

$$\tilde{S}_i = \mathcal{O}\left[\left(\frac{d_\varphi}{\epsilon^2} + \frac{d}{|\mathcal{N}|\, \epsilon^2} \right) \log \frac{RL_f L_h (RL_v + \rho_i s)}{\delta \epsilon} \right]. \tag{17}$$

Proof 3.1: See Appendix.

According to (17), The generalization performance of the proposed algorithm is influenced by the global model, sparse control units, and communication quality, because these parameters collectively determine the size of the parameter space that can be explored. Additionally, Theorem 3.1 further illustrates that as the sparsity level of the local model increases, the lower bound $\tilde{S}_i$ decreases, indicating that the algorithm exhibits better generalization performance.

Next, we investigate the computational and communication efficiency of the proposed algorithm. According to the communication quality of clients, pFedCAS is capable of adaptively sparsifying local models, which reduces computation and communication costs while improving the training accuracy of the entire system. As mentioned in Section II, the size of parameter φ_i of the sparse control unit is $d_\varphi = 2Bd_x$, which satisfies $d_\varphi \ll d$. We define $\overline{\rho}$ as the average communication quality factor of the entire system, which is given by $\overline{\rho} = \sum_i \rho_i / C$. Therefore, we can obtain the average computation cost of clients in each training epoch as $\mathcal{O}(\overline{\rho}\beta d + d_\varphi)$, while the average communication cost is $\mathcal{O}(\overline{\rho}\beta d)$. Moreover, since zero blocks are not

Algorithm 2: Personalized FL with Adaptive Sparse Control Unit in Local Training.

Required: $T, \eta_g, \eta_c, \beta, \beta_{\min}, \theta_g^0, \{\varphi_i^0\}_{i\in\mathcal{N}}$,
Initialize the global model h_{θ_g} with θ_g^0,
Initialize the N local models $\{h_{\theta_i}\}_{i\in\mathcal{N}}$ with $\{\varphi_i^0\}_{i\in\mathcal{N}}$,
for t $= 1 \to T$ **do**
  Initialize $\mathcal{N}_k$ as an empty set,
  for k $= 1 \to N$ **do**
    Calculate data rate R_k^t and get communication quality factor ρ_k,
    if $\rho_k s \geq s_{\min}$ **then**
      Add client k to the set $\mathcal{N}_w$,
    end if
  end for
  Sample partial clients $\mathcal{N}_a$ from $\mathcal{N}_w$,
  Sever broadcasts shared global model θ_g to clients $\mathcal{N}_a$,
  for client $i \in \mathcal{N}_a$ in parallel **do**
    for j $= 1 \to B$ **do**
      Get sparse control weight: $M'_{i,j}$,
      Get sparse local model: $\hat{\theta}_{i,j} = \theta_g^{t-1} \circledcirc, M'_{i,j}$,
      Calculate the loss function: $f(\hat{\theta}_{i,j}; \mathbf{x}_{i,j}, \mathbf{y}_{i,j})$,
      Update the sparse control unit:
      $\varphi_i^t = \varphi_i^{t-1} - \eta_c \nabla_\varphi \hat{\theta}_{i,j} \nabla_\theta f(\hat{\theta}_{i,j}; \mathbf{x}_{i,j}, \mathbf{y}_{i,j})$,
      Get sparse $\Delta\theta_{g,i}^t = -\eta_g \nabla_\theta f(\hat{\theta}_{i,j}; \mathbf{x}_{i,j}, \mathbf{y}_{i,j})$,
    end for
    Upload sparse $\Delta\theta_{g,i}^t$,
  end for
  Update global model via aggregating $\{\Delta\theta_{g,i}^t\}_{i\in\mathcal{N}}$.
end for

uploaded, the communication cost can be further reduced. pFedCAS can significantly reduce computation and communication costs while ensuring efficient training. Therefore, pFedCAS can effectively improve the timeliness of processing massive medical data in SHSs.

IV. Simulation Results and Analysis

In this section, we evaluate the accuracy, efficiency, and robustness of the proposed method to non-iid data through several simulation experiments. To fit the requirements of the SHS scenario, we utilized the HAM10000 dataset as the training and validation set for the simulation experiments.

A. Simulation Setting

The HAM10000 dataset used in the simulation experiments contains 10015 unique skin disease images, including seven common skin diseases. Each image has a size of $28 \times 28 \times 3$. Meanwhile, to ensure the non-iid distribution of local data among different clients, we define Dirichlet allocation factor $\alpha \in (0, 1]$, which represents the data heterogeneity [24]. The smaller α suggests a more substantial data heterogeneity.

In the simulation experiments, we configured 100 clients and specified the number of training rounds T as 400. Besides,

TABLE I
Simulation Parameter Setting

Parameters	Value
Noise power density N_0	-175dB m/Hz
Channel bandwidth W_i	180 kHz
Transmission power P_i	20 dBm
Channel gain g_i	$g_i \sim Uniform(20, 60)$ dB
Standard data rate R_{std}	6.5 MB
Minimum sparsity constraint β_{min}	0.2
Input feature size d_x	$28 \times 28 \times 3$
Batch size B	128
Learning rate of global model η_g	0.05($t < 300$), 0.005($t \geq 300$)
Learning rate of sparse control unit η_c	0.05($t < 300$), 0.005($t \geq 300$)
Size of Convolutional layer $Conv_1$, $Conv_2$	(3, 32, 5), (32, 64, 5)
Size of Pool layer	(2, 2)
Size of Fully connected layer Fc_1, Fc_2	(1024, 120), (120, 84)
Size of Output layer	(84, 7)

to improve the efficiency in the early stage of model training while ensuring convergence in the later stage, pFedCAS utilize a phased learning rate as shown in Table I. A summary of the remaining simulation parameter settings is provided in Table I. As mentioned in Section III, LeNet is employed as the neural network architecture of the global model, whose parameter configurations are presented in Table I.

B. Comparison of Different FL Algorithms

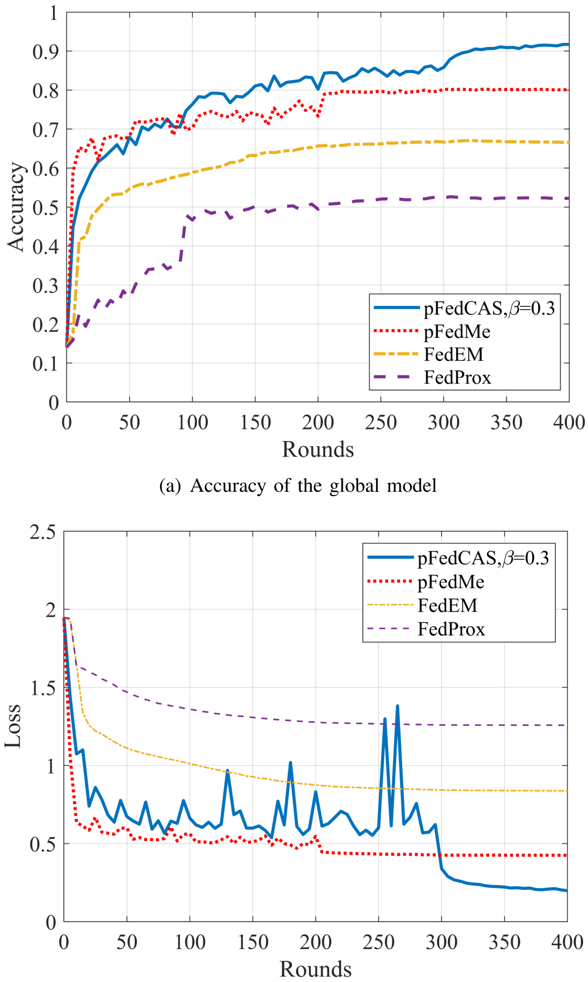
Firstly, we analyze the performance of pFedCAS compared with three baseline algorithms, namely pFedMe, FedEM and FedProx [25]. Specifically, we evaluate the accuracy and loss of the global model by applying four methods. In the comparative experiment, we set up Dirichlet allocation factor $\alpha = 0.4$, and sparse factor of the entire system $\beta = 0.3$. The selection of $\beta = 0.3$ is made to validate whether pFedCAS with a higher sparsity can guarantee a high model accuracy.

As shown in Fig. 5, the proposed algorithm outperforms the baselines regarding accuracy and loss. Compared with pFedMe, the proposed algorithm can improve the accuracy by about 15%. In addition, from Fig. 5(b), it can be observed that before the training round of pFedCAS with $\beta = 0.3$ reaches 300, the oscillations of the loss curve are more violent. There are two main reasons for this phenomenon: 1) the learning rate in the early stage of training is relatively high; 2) the sparsity level of the entire system is relatively high. However, in the later stage of training, when the learning rate decreases, the proposed algorithm can converge quickly.

C. Analysis of Costs of a Training Round

To validate the efficiency of the proposed algorithm in terms of computation and communication, we compared the training cost of pFedCAS with other baseline algorithms in one round. Specifically, we compared the average training time and peak memory usage of different algorithms in one round.

As shown in Figs. 5(a) and 6, compared to pFedMe, pFedCAS improves the performance by about 15% while using 70.8%



(a) Accuracy of the global model

(b) Loss of the global model

Fig. 5. Performance of different algorithms.

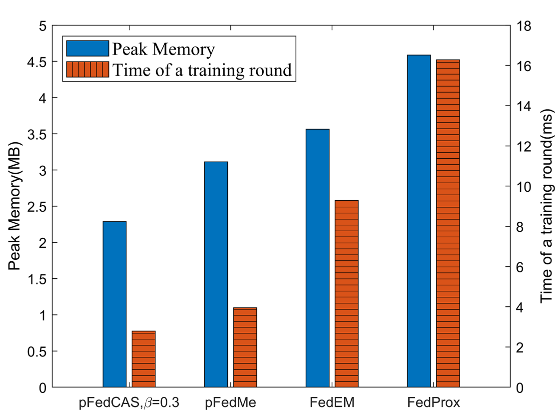
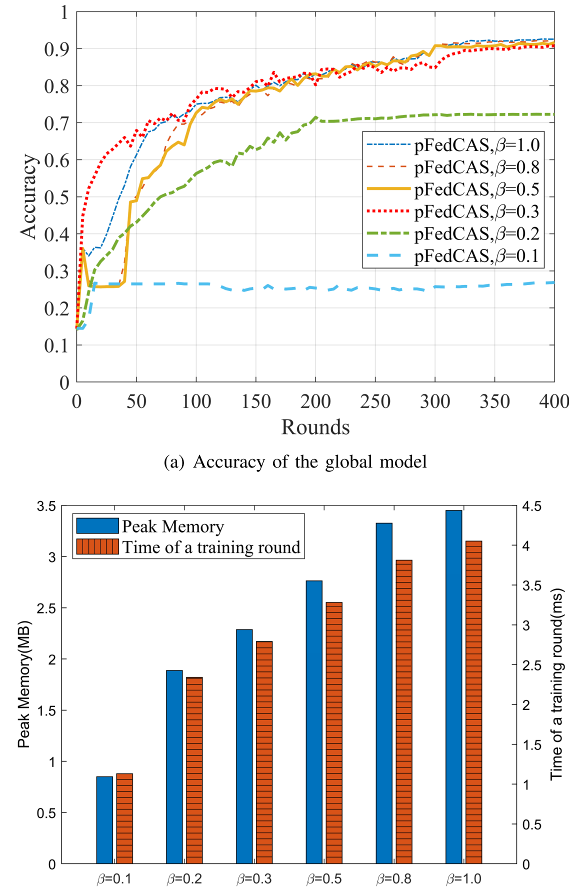


Fig. 6. Costs of a training round applying different algorithms.



(a) Accuracy of the global model

(b) Costs of a training round

Fig. 7. Performance of pFedCAS with different values of parameter β.

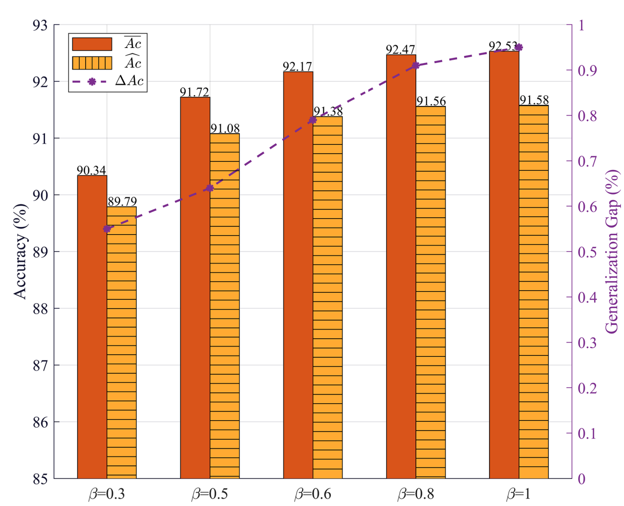
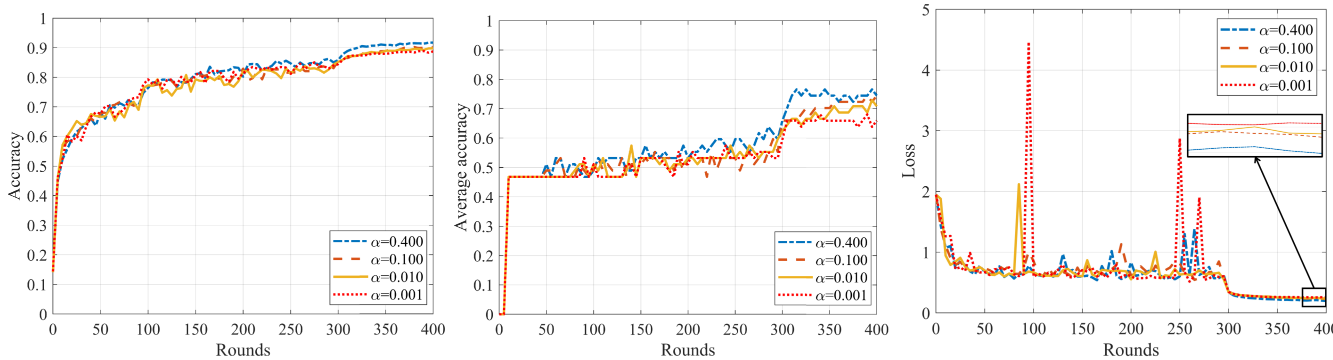


Fig. 8. Generalization gap ΔAc at different β.

of the training time and 73.5% of the memory. Compared to FedProx, pFedCAS achieved a performance improvement of 75.6% with only 17.1% of the training time and 50.0% of the memory. This is because pFedCAS significantly reduces the cost of model training by sparsifying the model, and sparsification only incurs minimal performance loss. In summary, compared to other baseline algorithms, pFedCAS can achieve better model performance with lower communication and computation costs.

D. Effect of Sparsity Factor β

Sparsity factor β, as the core parameter in the proposed algorithm, controls the sparsity level of the entire system. We



(a) Accuracy of the global model (b) Accuracy of the bottom 20% local models (c) Loss of the global model

Fig. 9. Performance of pFedCAS with different values of parameter α.

explored the impact of parameter β on the performance of the proposed algorithm by adjusting its values. It should be noted that in the simulation experiments, $\beta = 1$ represents that all clients do not perform sparsification, which reflects the optimal performance of the proposed algorithm without considering training costs.

As shown in Fig. 7, the performance of the global model degrades as the sparsity factor β decreases, where a decrease in β represents an increase in the sparsity level of the system. Specifically, when $\beta \geq 0.3$, the degradation of global model performance due to the increase in sparsity level is minimal; while when $\beta < 0.3$, an increase in sparsity level leads to a sharp drop in the global model performance. On one hand, excessive sparsity makes it difficult for local models to upload sufficient parameter updates. On the other hand, a small value of β indicates poor communication quality, resulting in only a few local models being selected to upload parameters. Therefore, it can be inferred that there exists a critical value $\beta_c \in (0.2, 0.3]$ for β, which can precisely lower the training costs while ensuring model performance. Besides, Fig. 6 intuitively reflects that an increase in sparsity level can lower the cost of model training. This is because the sparsity of local models not only significantly reduces the size of the parameters required for local model training but also reduces the number of parameter updates needed to be uploaded. In summary, under the premise that the sparsity level does not exceed β_c, we can adjust the sparsity factor β to reduce the costs of communication and computation while ensuring model performance.

E. Robustness to Non-Iid Data

Finally, under the sparsity factor $\beta = 0.3$, we vary the value of α, which represents the data heterogeneity level, to validate the robustness of the proposed algorithm to non-iid data. As shown in Fig. 9(a), the enhancement of data heterogeneity has little effect on the final performance of the global model. When α is reduced from 0.4 to 0.001, the accuracy of the global model only decreases by about 3%. However, Fig. 9(b) indicates that, compared to the global model, the enhancement of data heterogeneity may lead to a specific degradation in the performance of local models of the bottom 20% clients. This is because data heterogeneity may make it more difficult for local models to capture the characteristics of global data. Furthermore, as shown in Fig. 9(c), the enhancement of data heterogeneity may increase the oscillation amplitude of the loss curve of the global model, but it does not affect the final performance. This is because when local models of clients are used to update the global model, inconsistency and inaccuracy may be introduced, which increases the difficulty of model training. However, the proposed algorithm can greatly alleviate the performance loss caused by data heterogeneity.

F. Generalization Analysis

To evaluate the generalization performance of the proposed algorithm, we select 25% of the clients to not participate in FL training. After the model training is completed, the model is transferred to the non-participating clients, which only need to fine-tune the trained model using their local dataset. Finally, the accuracy of the participating clients and the non-participating clients is evaluated, denoted as $\overline{Ac}$ and $\widehat{Ac}$, respectively. Additionally, $\Delta Ac = \overline{Ac} - \widehat{Ac}$ is used to indicate the generalization gap of the algorithm.

As shown in Fig. 8, it can be observed that as the model sparsity level increases, the generalization gap of the model decreases. This is because the non-participating clients only need to train lightweight sparse control units without retraining the globally large parameterized model. This simulation result confirms Theorem 3.1, which indicates that model sparsity can enhance the generalization performance of pFedCAS.

V. Conclusion

To meet the requirements of privacy protection of healthcare data and high efficiency in SHSs, we proposed a pFedCAS framework to improve the privacy protection while ensuring communication efficiency. By introducing a sparsity control unit into the local model training, this framework could adaptively select the optimal sparsity factor to control the sparsity of the local model based on the communication quality of the SHS device. In addition, a selection unit was added during the global model aggregation to select local models that are suitable for parameter updates, thereby further improving the communication efficiency. We utilized the HAM10000 skin disease dataset to test the accuracy, efficiency, and robustness to non-iid data of the proposed method. Simulation experiments showed that compared with the baseline algorithm, the proposed method could not only achieves better privacy protection but also improves the model accuracy by 15% and reduces training costs by 30%. In addition, the proposed method also exhibited the excellent robustness to heterogeneous data distributions. After increasing the heterogeneity of the local data distribution by 100 times, the model accuracy only decreases by about 3%.

Appendix

Proof of Proposition 2.1

We can derive the following inequality from the assumption that f is μ-strongly convex:

$$f(\theta_1) - f(\theta_2) - \nabla_\theta f^\top(\theta_2)(\theta_1 - \theta_2) \geq \frac{\mu}{2}\left\|\theta_1 - \theta_2\right\|^2, \forall\theta_1,\theta_2. \tag{18}$$

Due to the arbitrariness of θ_1 and θ_2, we can exchange their positions in (18), and add the new inequality correspondingly to the original formula. We can obtain:

$$(\nabla_\theta f(\theta_1) - \nabla_\theta f(\theta_2))^\top(\theta_1 - \theta_2) \geq \mu\left\|\theta_1 - \theta_2\right\|^2, \forall\theta_1,\theta_2. \tag{19}$$

From Cauchy-Schwartz Inequality, we can obtain:

$$\begin{aligned}&\left\|\nabla_\theta f(\theta_1) - \nabla_\theta f(\theta_2)\right\| \quad \left\|\theta_1 - \theta_2\right\| \\ &\geq (\nabla_\theta f(\theta_1) - \nabla_\theta f(\theta_2))^\top(\theta_1 - \theta_2).\end{aligned} \tag{20}$$

Incorporating (19) and (20), we can get:

$$\left\|\nabla_\theta f(\theta_1) - \nabla_\theta f(\theta_2)\right\| \geq \mu\left\|\theta_1 - \theta_2\right\|, \forall\theta_1,\theta_2. \tag{21}$$

Under Assumption 2.1, substitute (θ_i^*, θ_g^*) to (21) leads to:

$$\left\|\theta_i^* - \theta_g^*\right\| \leq \frac{\sigma_i}{\mu}. \tag{22}$$

Replace θ_i^* with $M_i^* \circ \theta_g^*$ in (22), and we can get:

$$\left\|M_i^* \circ \theta_g^* - \theta_g^*\right\| = \left\|(M_i^* - 1) \circ \theta_g^*\right\| \leq \frac{\sigma_i}{\mu}. \tag{23}$$

Utilizing the Pólya-Szegö's Inequality, we can get:

$$\left\|(M_i^* - 1)^{\circ 2}\right\|^2 \left\|(\theta_g^*)^{\circ 2}\right\|^2 \leq A_i^2 \left\|(M_i^* - 1) \circ \theta_g^*\right\|^4. \tag{24}$$

According to (24), we can further derive Proposition 2.1 and finish the proof.

Proof of Theorem 3.1

We define $\mathcal{F}^{|\mathcal{N}|}$ as a function space, of which the size is $|\mathcal{N}|$. Then, the parameters of $\mathcal{F}^{|\mathcal{N}|}$ are defined as $(\theta_g, \varphi_1, \ldots, \varphi_{|\mathcal{N}|})$. Meanwhile, the distance function of $\mathcal{F}^{|\mathcal{N}|}$ is formulated as

$$\begin{aligned}&d[(\theta_g, \varphi_1, \ldots, \varphi_{|\mathcal{N}|}), (\theta'_g, \varphi'_1, \ldots, \varphi'_{|\mathcal{N}|})] = \\ &\mathbb{E}_{(x,y)\sim\mathcal{D}_i} \frac{\left|\sum f(\varphi_i, \theta_g; x, y)\right| - \left|\sum f(\varphi'_i, \theta'_g; x, y)\right|}{|\mathcal{N}|},\end{aligned} \tag{25}$$

where $f(\varphi_i, \theta_g; x, y)$ can be expressed as $f(\varphi_i, \theta_g; x, y) \triangleq f((v_{\varphi_i(x)} \circledcirc \theta_g); x, y)$.

According to Assumption 3.1, we have:

$$\begin{aligned}&d[(\theta_g, \varphi_1, \ldots, \varphi_{|\mathcal{N}|}), (\theta'_g, \varphi'_1, \ldots, \varphi'_{|\mathcal{N}|})] \\ &\leq \frac{1}{|\mathcal{N}|}\sum \mathbb{E}_{(x,y)\sim\mathcal{D}_i}\left|f(\varphi_i, \theta_g; x, y) - f(\varphi'_i, \theta'_g; x, y)\right| \\ &\leq \sum \frac{L_f}{|\mathcal{N}|}\left\|\theta_g - \theta'_g\right\| \leq L_f\left\|h_{\theta_g,\varphi_i} - h_{\theta'_g,\varphi'_i}\right\| \\ &\leq L_f L_h\left\|M_i \circ \theta_g - M'_i \circ \theta'_g\right\| \\ &= L_f L_h\left\|M_i \circ \theta_g - M'_i \circ \theta_g + M'_i \circ \theta_g - M'_i \circ \theta'_g\right\| \\ &\leq L_f L_h(\left\|M_i - M'_i\right\|\left\|\theta_g\right\| + \left\|M'_i\right\|\left\|\theta_g - \theta'_g\right\|) \\ &\leq L_f L_h(R\left\|M_i - M'_i\right\| + \rho_i s\left\|\theta_g - \theta'_g\right\|) \\ &= L_f L_h(R\left\|v_{\varphi_i}(x) - v_{\varphi'_i}(x)\right\| + \rho_i s\left\|\theta_g - \theta'_g\right\|) \\ &\leq L_f L_h(RL_v\left\|\varphi_i - \varphi'_i\right\| + \rho_i s\left\|\theta_g - \theta'_g\right\|) \leq \epsilon.\end{aligned} \tag{26}$$

According to (26), an ϵ-covering in space $\mathcal{F}^{|\mathcal{N}|}$ can be obtained when $\max(\|\varphi_i - \varphi'_i\|, \|\theta_g - \theta'_g\|) \leq \frac{\epsilon}{L_f L_h(RL_v + \rho_i s)}$. Therefore, let $\mathcal{C}(\mathcal{F}^{|\mathcal{N}|}, \epsilon)$ denote the covering size of $\mathcal{F}^{|\mathcal{N}|}$, which can be formulated as

$$\log\mathcal{C}(\mathcal{F}^{|\mathcal{N}|}, \epsilon) = \mathcal{O}[(|\mathcal{N}|\, d_\varphi + d)\log\frac{RL_f L_h(RL_v + \rho_i s)}{\epsilon}]. \tag{27}$$

According to the work [22], there exits a lower bound $\tilde{S}_i = \mathcal{O}[(\frac{d_\varphi}{\epsilon^2} + \frac{d}{|\mathcal{N}|\epsilon^2})\log\frac{RL_f L_h(RL_v + \rho_i s)}{\delta\epsilon}]$.

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